

Linear Algebra

Orthogonality

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01

Orthogonality

Orthogonal Sets

Definition

- A set of vectors $\{a_1, \dots, a_k\}$ in R^n is orthogonal set if each pair of distinct vectors is orthogonal (mutually orthogonal vectors).

A basis B of an inner product space V is called an **orthonormal basis** of V if

a) $\langle \mathbf{v}, \mathbf{w} \rangle = 0$ for all $\mathbf{v} \neq \boldsymbol{w} \in B$, and (mutual orthogonality)

b) $\|\mathbf{v}\| = 1$ for all $\mathbf{v} \in B$. (normalization)

- set of n−vectors $a_1, \dots, a_k$ are *(mutually) orthogonal* if $a_i \perp a_j$ for $i \neq j$
- They are *normalized* if $\|a_i\| = 1$ for $i = 1, \dots, k$
- They are *orthonormal* if both hold
- Can be expressed using inner products as $a_i^T a_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

Orthogonal Sets

- Geometry



https://youtu.be/dqdSzqsm7bY

- Algebra

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = \mathbf{0}$.

Suppose V is an inner product space.
Two vectors $\mathbf{v}, \mathbf{w} \in V$ are called **orthogonal** if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$.

The Pythagorean Theorem

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$

Orthogonal Sets

Theorem (1)

If $S = \{a_1, \dots, a_k\}$ is an orthogonal set of nonzero vectors in R^n, then S is linearly independent and is a basis for the subspace spanned by S.

Proof

If k = n, then prove that S is a basis for R^n

Linear combinations of orthonormal vectors

Corollary

- A simple way to check if an n−vector y is a linear combination of the orthonormal vectors $a_1, \dots, a_k$, if and only if:

$$y = (a_1^T y)a_1 + \dots + (a_k^T y)a_k$$

- For orthogonal vectors $a_1, \dots, a_k$:

$$y = c_1 a_1 + \cdots + c_k a_k$$

$$c_j = \frac{y.a_j}{a_j.a_j}$$

Orthonormal vectors

- Orthonormal sets of vectors are linearly independent
- By independence-dimension inequality, must have $k \leq n$
- When $k = n, a_1, \ldots, a_n$ are an *orthonormal basis*

Review "Independence–dimension inequality"

If the n–vectors $a_1, \ldots, a_k$ are linearly independent, then $k \leq n$.

Linear combinations of orthonormal vectors

Example

Write x as a linear combination of a_1, a_2, a_3?

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad a_1 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad a_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad a_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

02

Orthogonal Subspaces

Definition

- Two subspaces W_1 and W_2 of the same space V are orthogonal, denoted by $W_1 \perp W_2$, if and only if each vector $w_1 \in W_1$ is orthogonal to each vector $w_2 \in W_2$ for all w_1, w_2 in W_1, W_2 respectively:

$$< w_1, w_2 > = 0$$

Example

If the bases of two subspaces are orthogonal, it implies that the subspaces themselves are orthogonal.

03

Orthogonal Complements

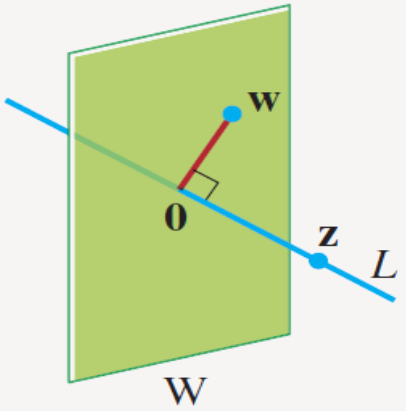
Definition

- If a vector z is orthogonal to every vector in a subspace W of R^n, then z is said to be orthogonal to W.
- **The set of all vectors z that are orthogonal** to W is called the orthogonal complement of W and is denoted by $W^{\perp}$

Example

W be a plane through the origin in $\mathbb{R}^3$.

$L = W^{\perp}$ and $W = L^{\perp}$



Orthogonal Complements

Theorem (2)

$W^{\perp}$ is a subspace of $\mathbb{R}^n$.

Theorem (3)

$W^{\perp} \cap W = \{\mathbf{0}\}$.

Important

We emphasize that W_1 and W_2 can be orthogonal without being complements.

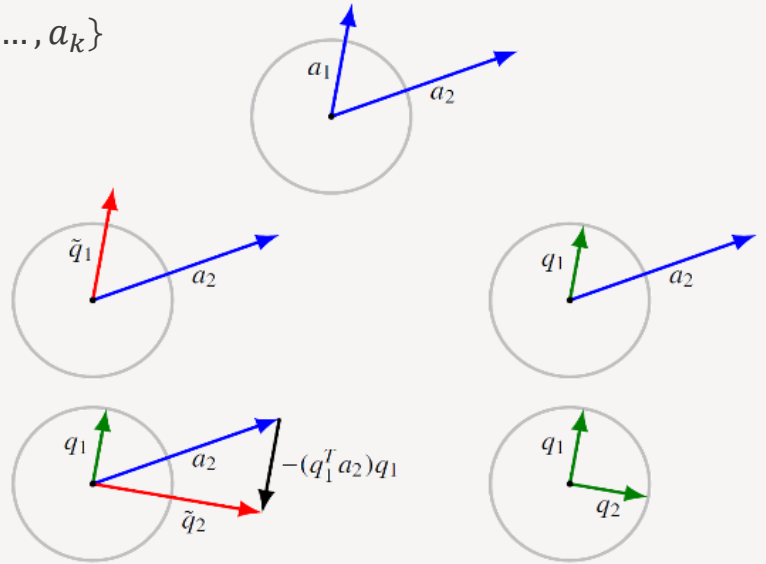
$W_1 = span((1,0,0))$ and $W_2 = span((0,1,0))$.

04

Gram–Schmidt Algorithm

Gram–Schmidt (orthogonalization) algorithm

- Find orthonormal basis for span $\{a_1, a_2, \dots, a_k\}$
- Geometry:



Gram–Schmidt (orthogonalization) algorithm

- Find orthonormal basis for span $\{a_1, a_2, \dots, a_k\}$
- Algebra:

1) $q1 = \frac{a_1}{\|a_1\|}$

2) $\widetilde{q_2} = a_2 - (q_1^T a_2) q_1 \rightarrow q_2 = \frac{\widetilde{q_2}}{\|\widetilde{q_2}\|}$

3) $\widetilde{q_3} = a_3 - (q_1^T a_3) q_1 - (q_2^T a_3) q_2 \rightarrow q_3 = \frac{\widetilde{q_3}}{\|\widetilde{q_3}\|}$

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k) $\widetilde{q_k} = a_k - (q_1^T a_k) q_1 - \cdots - (q_{k-1}^T a_k) q_{k-1} \rightarrow q_k = \frac{\widetilde{q_k}}{\|\widetilde{q_k}\|}$

Gram–Schmidt (orthogonalization) algorithm

Example

Find orthogonal set for $a = \begin{bmatrix}1\\0\\1\end{bmatrix}, b = \begin{bmatrix}1\\0\\0\end{bmatrix}, \mathrm{c} = \begin{bmatrix}2\\1\\0\end{bmatrix}$

Gram–Schmidt (orthogonalization) algorithm

- Why $\{q_1, q_2, \dots, q_k\}$ is an orthonormal basis for span $\{a_1, a_2, \dots, a_k\}$?
  - $\{q_1, q_2, \dots, q_k\}$ are normalized.
  - $\{q_1, q_2, \dots, q_k\}$ is an orthogonal set
  - a_i is a linear combination of $\{q_1, q_2, \dots, q_i\}$

  span$\{q_1, q_2, \dots, q_k\}$ = span$\{a_1, a_2, \dots, a_k\}$

- q_i is a linear combination of $\{a_1, a_2, \dots, a_i\}$

Gram–Schmidt (orthogonalization) algorithm

- Given n-vectors $a_1, \dots, a_k$ for $i = 1, \dots, k$

  1. Orthogonalization: $\tilde{q}_i = a_i - (q_1^T a_i)q_1 - \cdots - (q_{i-1}^T a_i)q_{i-1}$
  2. Test for linear dependence: if $\tilde{q}_i = 0$, quit
  3. Normalization: $q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}$

Note

- If G–S does not stop early (in step 2), $a_1, \dots, a_k$ are linearly independent.
- If G–S stops early in iteration $i = j$, then a_j is a linear combination of $a_1, \dots, a_{j-1}$ (so $a_1, \dots, a_k$ are linearly dependent)

$$a_j = (q_1^T a_j)q_1 + \cdots + (q_{j-1}^T a_j)q_{j-1}$$

Complexity of Gram–Schmidt algorithm

- Gram-Schmidt algorithm gives us an explicit method for determining if a list of vectors is linearly dependent or independent.
- What is complexity and number of flops for this algorithm?
  - $O(nk^2)$ why?
- Given n-vectors $a_1, \dots, a_k$ for $i = 1, \dots, k$

1. Orthogonalization: $\tilde{q}_i = a_i - (q_1^T a_i)q_1 - \cdots - \left(q_{i-1}^T a_i\right)q_{i-1}$
2. Test for linear dependence: if $\tilde{q}_i = 0$, quit
3. Normalization: $q_i = \frac{\tilde{q}_i}{\|\tilde{q}_i\|}$

Orthonormal basis

Corollary

Every finite-dimensional inner product space has an orthonormal basis.

Conclusion

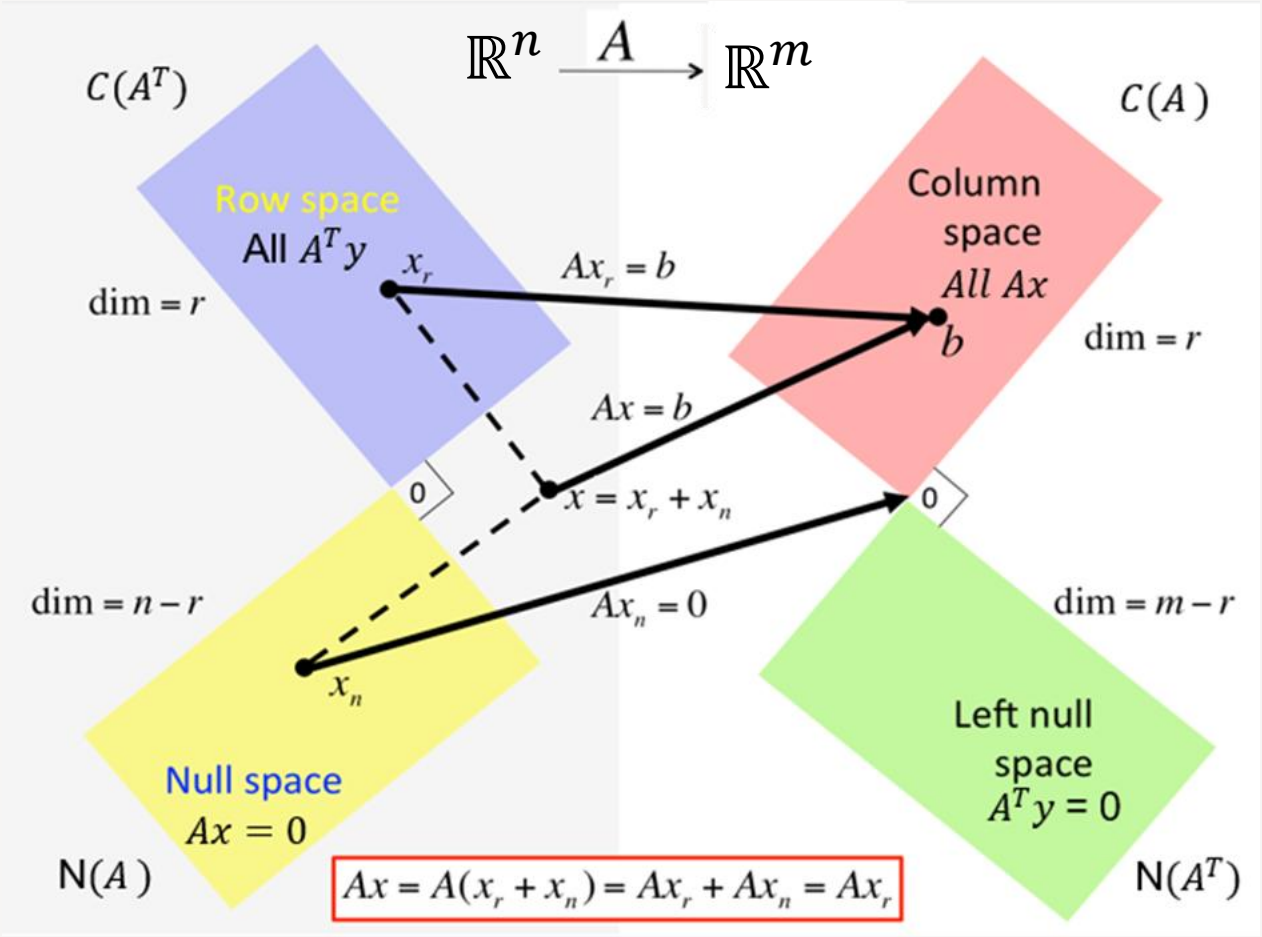
Existence of Orthonormal Bases

- Every finite-dimensional inner product space has an orthonormal basis.
- Since finite-dimensional inner product spaces (by definition) have a basis consisting of finitely many vectors, and the Gram-Schmidt process tells us how to convert that basis into an orthonormal basis, we now know that every finite-dimensional inner product space has an orthonormal basis.

05

Four Fundamental Subspaces

(of Matrix Space)


$\mathbb{R}^n$ A $\mathbb{R}^m$
$C(A^T)$
$C(A)$
Row space
All $A^T y$
Column space
All Ax
x_r
$Ax_r = b$
b
dim = r
dim = r
$Ax = b$
$x = x_r + x_n$
0
0
dim = $n - r$
$Ax_n = 0$
dim = $m - r$
x_n
Null space
$Ax = 0$
Left null space
$A^T y = 0$
$N(A)$
$N(A^T)$
$Ax = A(x_r + x_n) = Ax_r + Ax_n = Ax_r$

References

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- Chapter 6: Linear Algebra David Cherney
- Linear Algebra and Optimization for Machine Learning
- Introduction to Applied Linear Algebra Vectors, Matrices, and Least Squares